

SRS Section 9 — Non-Functional and Operational System Requirements

📌 Draft v0.1 / pre-baseline. This page is **section 9 of the [Software Requirements Specification](#)** and is governed by that document's status, evidence-authority hierarchy and status conventions. It is held as a child page for size and editability only; it is not a separate artifact and carries no independent approval. The controlled business source is the [Non-Functional and Operational Requirements](#) page, which remains a separate supporting page and is not merged, altered or superseded by this content. Publication creates no approval, service-level commitment, implementation acceptance, UAT acceptance or release authorisation.

9.1 How to read this section

Each requirement carries the identifier `SNFR-nnn` and restates a business non-functional or operational requirement as a testable system statement. Source NFR requirements NFR-01 to NFR-16 are neither rewritten nor replaced; their wording remains owned by the NFR page.

⚠️ No numerical target in this section is an approved service level. Where the NFR page records a proposed figure, that figure is reproduced here *as a proposal attributed to its source* and its approval state is stated explicitly. Where no figure exists, the target is recorded as **TBD — requires validation** with a named owner role and the gate by which the decision is required. No target has been invented, rounded, inferred from architecture, or carried over from another project.

Target approval states used in this section:

State	Meaning
EVIDENCED	The requirement is met and verification evidence exists in the canonical repository. Applies to design-time properties only; no runtime service level holds this state.
PROPOSED — NOT APPROVED	A figure exists on the NFR page as a proposal. It has no accountable-owner approval and may not be cited as a commitment.
TBD — REQUIRES VALIDATION	No target exists. The named owner must set one before the stated gate.

Environments referenced are those in the NFR page operational model: development, test, pre-production and production. Where a requirement can only be evidenced in an environment that does not yet exist, that is stated rather than glossed.

9.2 SNFR index

SNFR	Quality attribute	Source	Target approval state	Status — Catalogue / RTM
SNFR-001	Grounding and numeric correctness	NFR-01	Evidenced (zero tolerance)	Implemented / Designed-not implemented
SNFR-002	Determinism and versioning	NFR-02	Evidenced	Implemented / Designed-not implemented
SNFR-003	Least privilege and RBAC	NFR-03	Partly evidenced; production IAM TBD	Designed-not implemented / Designed-not implemented
SNFR-004	Privacy — aggregate-only executive scope	NFR-04	Partly evidenced; UI paths TBD	Designed-not implemented / Partly implemented
SNFR-005	Auditability	NFR-05	TBD — requires validation	Designed-not implemented / Designed-not implemented
SNFR-006	Data integrity and rollback	NFR-06	Evidenced (checksum equality)	Implemented / Implemented
SNFR-007	Availability	NFR-07	Proposed — not approved	Designed-not implemented / Designed-not implemented

SNFR-008	Data residency	NFR-08	TBD — requires validation	Designed—not implemented / Designed—not implemented
SNFR-009	Performance — governed question response	NFR-09	Proposed — not approved	Designed—not implemented / Designed—not implemented
SNFR-010	Performance — non-AI interaction	NFR-10	Proposed — not approved	Designed—not implemented / Designed—not implemented
SNFR-011	Capacity and scalability	NFR-11	Proposed — not approved	Designed—not implemented / Designed—not implemented
SNFR-012	Supportability	NFR-12	Proposed — not approved	Designed—not implemented / Designed—not implemented
SNFR-013	Backup, recovery and continuity	NFR-13	Proposed — not approved	Designed—not implemented / Designed—not implemented
SNFR-014	Retention and legal hold	NFR-14	TBD — requires validation	Designed—not implemented / Designed—not implemented
SNFR-015	Accessibility	NFR-15	Standard named; audit not executed	Designed—not implemented / Designed—not implemented

SNFR-016	Usability — viewport and operability	NFR-16	Partly proposed; minimum viewport TBD	Designed—not implemented / Designed—not implemented
SNFR-017	Portability — dataset independence	NO SO...	Evidenced	Implemented / Implemented
SNFR-018	Observability and monitoring	NFR page, operational model	TBD — requires validation	Designed—not implemented / Designed—not implemented
SNFR-019	Maintainability and change control	NFR page, operational model	Partly evidenced	Partly implemented / Designed—not implemented
SNFR-020	Data quality	NO SO...	TBD — requires validation	Partly implemented / Designed—not implemented

❌ **Catalogue coverage gap identified during Stage SRS-3.** Two mandated quality attributes have **no requirement in the NFR catalogue: portability** (SNFR-017) and **data quality** (SNFR-020). Both are recorded here as derived requirements with their source marked *No source NFR* so the gap remains visible rather than being closed silently by this specification. Portability is a delivered and evidenced capability with no business requirement behind it; data quality is an undelivered expectation with no business requirement behind it. Neither may be treated as an approved requirement until added to the NFR catalogue under Change Control. This finding does not alter the NFR count, which remains 16.

9.3 Correctness and determinism

SNFR-001 — Grounding and numeric correctness

Source. NFR-01. **Related.** FR-04, BR-05; SFR-010, SFR-011; UC-02, UC-03, UC-05; UAT-07.

Shall. The system shall display no numeric statement that does not trace to a retained trusted tool result, and every blocked statement shall produce a retrievable trace.

Measurement method. Count of displayed unsupported numeric statements across the retained grounding test set and the golden-set evaluation, plus inspection of retained blocked outputs.

Threshold or target. Zero unsupported numeric statements displayed. This is a zero-tolerance boundary, not a percentage target, and admits no acceptable failure rate.

Target approval state. EVIDENCED at the backend for the retained test set. The zero-tolerance position is stated on the NFR page and requires no numerical validation.

Environment. Development and test today; pre-production once an interface exists.

Evidence required. Retained grounding test set results; golden-set evaluation records; retained blocked outputs resolvable by trace identifier; a complete post-fix live run.

Owner role. Product Owner.

Failure or exception treatment. Fail closed. The whole response is withheld rather than partially released. No exception, waiver or configuration may relax this; any reduction requires explicit executive, product, compliance and security approval under Change Control, and no prompt-only workaround is acceptable.

Implementation status. Catalogue: **Implemented** · RTM: **Designed / not implemented**.

Backend validation verified; no display surface exists to which it applies end to end.

Open items. The complete post-fix live evidence is incomplete under R-02 and DEP-08; GQ-038 has never passed live.

SNFR-002 — Determinism and versioning

Source. NFR-02. **Related.** FR-02, FR-13; SFR-004, SFR-005, SFR-033; UC-01, UC-05, UC-08.

Shall. The system shall produce identical outputs for identical inputs under identical component versions for compliance verdicts, ranking inputs, data validation, truth generation and chart validation, and shall retain the versions in force.

Measurement method. Repeated execution with fixed inputs and fixed versions, compared byte-for-byte; empty-diff regression gate on regenerated artefacts and truths.

Threshold or target. Identical output on repeat execution; empty diff where no input changed.

Target approval state. EVIDENCED for the components within the current backend scope.

Environment. Development and test.

Evidence required. Repeat-execution comparison records; retained diff reports including empty diffs; recorded ruleset, model and configuration versions.

Owner role. Engineering/Delivery Lead.

Failure or exception treatment. A non-empty unexplained diff blocks acceptance of the affected generation and triggers rollback under SNFR-006.

Implementation status. Catalogue: **Implemented** · RTM: **Designed / not implemented**.

Open items. Determinism of language-model narration is out of scope for this requirement by design; only computed values are covered.

9.4 Security, privacy and audit

SNFR-003 — Least privilege and role-based access control

Source. NFR-03. **Related.** FR-05, SEC-04; SFR-012, SFR-014, SFR-037; UC-02, UC-04, UC-05, UC-08, UC-09; UAT-02.

Shall. The system shall enforce least privilege server-side for every tool, route, dataset and export path, and prompt text shall not constitute a security boundary.

Measurement method. Role-negative test matrix covering every persona against every capability; inspection of identity and access management configuration once it exists.

Threshold or target. Zero unauthorised capability grants across the role matrix; 100 per cent of matrix cells exercised.

Target approval state. Backend enforcement EVIDENCED; production identity and access management, single sign-on and multi-factor authentication remain TBD — REQUIRES VALIDATION by the Information Security and Privacy Owner before UAT entry.

Environment. Test today; pre-production and production once identity integration exists.

Evidence required. Role-matrix test results; production IAM and SSO configuration evidence; joiner–mover–leaver procedure; quarterly access review with privileged access reviewed monthly.

Owner role. Information Security and Privacy Owner.

Failure or exception treatment. Any unauthorised grant is a release-blocking defect. Reduction of RBAC requires explicit executive, product, compliance and security approval.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**. Dispatcher enforcement is verified; production IAM is absent.

Open items. DEP-05 outstanding; role fit unvalidated under A-02.

Source. NFR-04. **Related.** BR-02, FR-05, SEC-05; SFR-013, SFR-039; UC-02, UC-03; UAT-02.

Shall. The system shall expose aggregates only to executive and viewer roles, and shall gate member and claim detail to authorised roles across every user interface, application programming interface and export path.

Measurement method. Role-negative tests executed separately against each access path — tool, route and export — rather than against the tool path alone.

Threshold or target. Zero record-level rows returned to a viewer session on any path.

Target approval state. Tool path **EVIDENCED**; interface and export paths **TBD — REQUIRES VALIDATION** by the Information Security and Privacy Owner before UAT entry, because those paths do not yet exist to be tested.

Environment. Test today; pre-production once interfaces exist.

Evidence required. Per-path role-negative results; export-scope evidence; retained denial records.

Owner role. Information Security and Privacy Owner.

Failure or exception treatment. Any record-level disclosure to a viewer is a privacy incident and follows the incident procedure, not a defect triage path.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Partly implemented**.

Open items. Export paths undeliverable until DEP-05; classification unresolved under I-06.

SNFR-005 — Auditability

Source. NFR-05. **Related.** FR-09, FR-11, FR-12, SEC-07; SFR-023, SFR-025, SFR-029, SFR-030; UC-07, UC-08; UAT-08, UAT-12.

Shall. The system shall record user, role, time, prompt, tool calls and parameters, result hashes, rules, model and configuration versions, response and approval events, append-only and searchable.

Measurement method. Field-completeness inspection across a sampled event set; search performance and correctness testing; negative testing of alteration and deletion attempts.

Threshold or target. 100 per cent of the mandated fields present on every event; zero successful alterations or deletions. Search response and retention-period targets are **TBD — requires validation**; no figure exists on the NFR page and none is invented here.

Target approval state. **TBD — REQUIRES VALIDATION** by the Compliance and Regulatory Owner, with the Platform/Support Owner for search and retention performance, before UAT entry.

Environment. Pre-production and production; not evidenceable today.

Evidence required. Sampled event records showing complete fields; retained rejection evidence for alteration attempts; audit-write failure monitoring evidence.

Owner role. Compliance and Regulatory Owner.

Failure or exception treatment. Audit-write failure is monitored and alerts. A period during which audit writes failed is a gap that must be declared, not reconstructed.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**. No persistent application audit service exists.

Open items. DEP-05 and DEP-06 outstanding; retention schedule unapproved under SNFR-014.

SNFR-008 — Data residency

Source. NFR-08. **Related.** SEC-02, SEC-03; all use cases. **Position in this section.** Recorded here with the security group because it is a control, not a performance attribute.

Shall. The system shall keep production health-data storage and model inference within the United Arab Emirates unless a documented lawful exception has been approved in writing before use.

Measurement method. Build-time verification of model routing configuration; data-flow and routing evidence covering prompts, outputs and support access; inspection of regional service configuration.

Threshold or target. Zero production health-data storage or inference outside the United Arab Emirates without an approved documented exception. Global and geographic model routing disabled for regulated production health data.

Target approval state. TBD — REQUIRES VALIDATION by the Information Security and Privacy Owner before UAT entry. Architecture intent exists; deployment evidence does not, and in-region model availability is unconfirmed.

Environment. Production; pre-production where regulated data is used.

Evidence required. Routing configuration evidence; data-flow and residency diagram including prompts, outputs and support access; model availability and terms evidence; any exception approval.

Owner role. Information Security and Privacy Owner.

Failure or exception treatment. Unverified routing blocks deployment. "UAE hosted" is insufficient where prompts or outputs may route outside the region.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**.

Open items. Risk R-03, assumption A-05 and dependency DEP-04 all open. Model availability, terms, subprocessors, logging and support access can change and require build-time re-verification.

9.5 Integrity, recoverability and portability

SNFR-006 — Data integrity and rollback

Source. NFR-06. **Related.** BR-06, FR-13; SFR-032 to SFR-035; UC-01; UAT-09.

Shall. On rejection of a data generation the system shall restore data, configuration, artefacts and golden truths to the previously accepted state.

Measurement method. Checksum comparison of all four component classes against the prior accepted state after a forced rejection; mutation-swap end-to-end test.

Threshold or target. Checksum equality across all four component classes; zero reachable mixed-version states.

Target approval state. EVIDENCED.

Environment. Development and test; repeatable in pre-production.

Evidence required. Rollback test results with checksum comparison; retained decision log for accepted and rejected generations.

Owner role. Data Owner/Steward.

Failure or exception treatment. Rollback failure raises an operational alert and blocks further onboarding until resolved.

Implementation status. Catalogue: **Implemented** · RTM: **Implemented**. Both sources agree.

Open items. None specific to this requirement.

SNFR-017 — Portability: dataset independence

 **Derived requirement — no source NFR exists.** Portability is a delivered, evidenced capability with no corresponding entry in the NFR catalogue. It is recorded here so the specification is complete and the catalogue gap stays visible. It is not an approved requirement until added to the NFR page under Change Control.

Source. None in the NFR catalogue. Derived from BR-06, FR-13 and the Gen-2 close-out scope.

Related. SFR-032 to SFR-035; UC-01; UAT-09.

Shall. The system shall onboard a conforming dataset through configuration alone, without code change, with configuration as the single source of truth for tool contracts, assembled prompts and view bindings.

Measurement method. Mutation-swap end-to-end onboarding of a substituted dataset; byte-identical regeneration of generated artefacts from configuration; empty-diff regression gate.

Threshold or target. Zero code changes required to onboard a conforming dataset; byte-identical artefact regeneration for an unchanged configuration.

Target approval state. EVIDENCED — subject to the caveat that no business requirement authorises this target.

Environment. Development and test.

Evidence required. Mutation-swap test results; configuration version and hash stamping on tool-result envelopes; onboarding runbook.

Owner role. Engineering/Delivery Lead, pending assignment of a business owner when the requirement is added to the catalogue.

Failure or exception treatment. A dataset requiring code change is out of the configuration-driven boundary and is not onboarded under this requirement. Inference of table roles, grain, relationships or column meanings remains prohibited under SFR-032 and SFR-049.

Implementation status. Catalogue: **Implemented** · RTM: **Implemented**.

Open items. The catalogue gap itself. Approved extracts, mappings and quality thresholds outstanding under DEP-03.

SNFR-013 — Backup, recovery and continuity

Source. NFR-13. **Related.** SNFR-006, SNFR-012; operational readiness.

Shall. The system shall meet approved recovery point and recovery time objectives and shall demonstrate restoration at an approved test frequency.

Measurement method. Restore test from backup measuring achieved recovery point and recovery time against the approved objectives.

Threshold or target. PROPOSED — NOT APPROVED: recovery point objective 24 hours and recovery time objective 8 hours for pre-production, as recorded on the NFR page. Restore test frequency is **TBD — requires validation**; no frequency is stated on the NFR page and none is invented here.

Target approval state. Proposed figures require validation by the Platform/Support Owner before requirements-baseline approval; restore evidence is required before UAT entry.

Environment. Pre-production; production before release.

Evidence required. Backup configuration; restore test evidence with measured achieved values; disaster-recovery test record.

Owner role. Platform/Support Owner.

Failure or exception treatment. An untested backup is treated as no backup. A restore test that misses the objective rebaselines the objective or the architecture, not the evidence.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**.

Open items. Assumption A-04 and dependency DEP-06 open.

9.6 Performance, capacity and availability

 Every figure in this subsection is a **proposal recorded on the NFR page**, reproduced with attribution. None has accountable-owner approval, none has been tested, and none may be quoted as a service level, a contractual commitment or an acceptance criterion. No load testing has been performed and no interface exists to load-test.

SNFR-007 — Availability

Source. NFR-07. **Related.** SNFR-012, SNFR-013, SNFR-018; operational readiness.

Shall. The pre-production service shall meet an approved availability target during approved support hours.

Measurement method. Monitoring of service availability across the agreed support window, measured over an agreed reporting period.

Threshold or target. PROPOSED — NOT APPROVED: 99.5 per cent during agreed support hours, as recorded on the NFR page. The support window itself is not yet defined, so the proposed figure is not currently measurable.

Target approval state. Requires validation by the Platform/Support Owner before requirements-baseline approval.

Environment. Pre-production.

Evidence required. Approved support window and availability target recorded before UAT; monitoring evidence showing the tested service meets the recorded value.

Owner role. Platform/Support Owner.

Failure or exception treatment. Planned maintenance treatment, exclusions and the measurement period are undefined and must be settled as part of approving the target.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**.

Open items. A-04 and DEP-06 open. No environment exists in which to measure this.

SNFR-009 — Performance: governed question response

Source. NFR-09. **Related.** FR-01, FR-04, FR-07, FR-08; SFR-001, SFR-010, SFR-018, SFR-020; UC-02, UC-03.

Shall. A governed aggregate question shall return a validated response within an approved percentile target, measured end to end including tool execution and grounding validation.

Measurement method. Percentile latency measurement over a representative question set under an agreed concurrency profile, measured from request receipt to validated response release.

Threshold or target. PROPOSED — NOT APPROVED: 95th percentile at or below 15 seconds including tools, as recorded on the NFR page. Unvalidated under load; the representative question set and concurrency profile are undefined.

Target approval state. Requires validation by the Platform/Support Owner before requirements-baseline approval, with load evidence before UAT entry.

Environment. Pre-production under an agreed load profile.

Evidence required. Load and performance test report stating the question set, concurrency profile, measurement boundary and achieved percentiles.

Owner role. Platform/Support Owner.

Failure or exception treatment. A grounding re-ask extends elapsed time; whether re-ask time counts toward the target is undefined and must be settled when the target is approved. This specification does not decide it.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**.

Open items. A-04 open. Model latency is partly outside the delivery team's control and depends on routing decisions under SNFR-008.

SNFR-010 — Performance: non-AI interaction

Source. NFR-10. **Related.** FR-07, FR-08, FR-09, FR-15, FR-16; SFR-017, SFR-021, SFR-023, SFR-038, SFR-040.

Shall. Non-AI page and application programming interface interactions shall meet an approved percentile target.

Measurement method. Percentile latency measurement across defined non-AI routes under an agreed concurrency profile.

Threshold or target. **PROPOSED — NOT APPROVED**: 95th percentile at or below 2 seconds, as recorded on the NFR page. The route set to which it applies is undefined.

Target approval state. Requires validation by the Platform/Support Owner before requirements-baseline approval.

Environment. Pre-production.

Evidence required. Load and performance test report naming the covered routes.

Owner role. Platform/Support Owner.

Failure or exception treatment. Undefined pending target approval.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**. No routes exist to measure.

Open items. A-04 and DEP-05 open.

SNFR-011 — Capacity and scalability

Source. NFR-11. **Related.** SNFR-009, SNFR-010; SFR-032, SFR-033.

Shall. The system shall support an approved number of named users, an approved peak concurrency, an approved extract size and an approved growth expectation.

Measurement method. Load testing at the approved concurrency; onboarding testing at the approved extract size; growth modelling against the approved expectation.

Threshold or target. **PROPOSED — NOT APPROVED**: a starting assumption of 50 named users and 10 concurrent, as recorded on the NFR page. Extract size and growth expectation are **TBD — requires validation**; no figures exist and none are invented here.

Target approval state. Requires validation by the Platform/Support Owner, with the Data Owner/Steward for extract size, before requirements-baseline approval.

Environment. Pre-production.

Evidence required. Load test at approved concurrency; onboarding test at approved extract size; documented growth assumption.

Owner role. Platform/Support Owner.

Failure or exception treatment. Behaviour beyond approved capacity — queueing, shedding or degradation — is undefined and must be specified when the capacity target is approved.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**.

Open items. A-04 and DEP-03 open.

9.7 Operability

SNFR-012 — Supportability

Source. NFR-12. **Related.** SNFR-007, SNFR-018.

Shall. The system shall be supported under approved business hours, severity definitions, escalation paths and incident service levels.

Measurement method. Incident response and resolution measurement against the approved severity matrix.

Threshold or target. PROPOSED — NOT APPROVED: United Arab Emirates business-hours support, as recorded on the NFR page. Severity definitions, escalation paths and incident service levels are **TBD — requires validation**; none exist and none are invented here.

Target approval state. Requires validation by the Platform/Support Owner before requirements-baseline approval.

Environment. Pre-production and production.

Evidence required. Support rota; severity matrix; escalation procedure; incident and breach-response plan.

Owner role. Platform/Support Owner.

Failure or exception treatment. Out-of-hours incident handling is undefined and must be settled with the support model.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**.

Open items. A-04 and DEP-06 open.

SNFR-018 — Observability and monitoring

Source. NFR page, operational model assumptions. **Related.** SNFR-005, SNFR-007, SNFR-012, SNFR-020; SFR-025.

Shall. The system shall monitor and alert on service health, failed grounding, denied access, data-generation failures, audit-write failures and abnormal export activity.

Measurement method. Alert catalogue completeness against the six named signal classes; synthetic fault injection confirming each alert fires.

Threshold or target. All six named signal classes monitored with a defined alert. Alert thresholds, notification routing and acknowledgement times are **TBD — requires validation**; none are stated on the NFR page and none are invented here.

Target approval state. **TBD — REQUIRES VALIDATION** by the Platform/Support Owner before UAT entry.

Environment. Pre-production and production.

Evidence required. Monitoring and alert catalogue; fault-injection evidence per signal class; on-call routing configuration.

Owner role. Platform/Support Owner.

Failure or exception treatment. An unmonitored signal class is a readiness gap, not an accepted risk. Audit-write failure monitoring is a control dependency of SNFR-005 and cannot be deferred separately from it.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**.

Open items. DEP-06 open. Monitoring appears only as an operational assumption on the NFR page rather than as a numbered requirement.

SNFR-019 — Maintainability and change control

Source. NFR page, operational model assumptions. **Related.** SNFR-002, SNFR-006; SFR-033, SFR-034.

Shall. Every change deployment shall require peer review, passing automated tests, a passing dry evaluation, a migration and rollback plan, and recorded approval.

Measurement method. Inspection of the deployment pipeline gates; sampled inspection of change records against the five required elements.

Threshold or target. All five elements present for every deployment; zero deployments bypassing a gate.

Target approval state. Test and dry-evaluation gates **EVIDENCED** in the current repository. Peer review, migration and rollback planning and recorded approval remain **TBD — requires validation** as a formalised procedure.

Environment. All environments.

Evidence required. Pipeline configuration; sampled change records; retained test and dry-evaluation results per change.

Owner role. Engineering/Delivery Lead.

Failure or exception treatment. An emergency change path, if any, must itself be defined and recorded; none currently exists.

Implementation status. Catalogue: **Partly implemented** · RTM: **Designed / not implemented**.

Open items. The delivery-sequencing deviation recorded as I-10 indicates that phase sequencing is not currently gated by a recorded approval step.

SNFR-014 — Retention and legal hold

Source. NFR-14. **Related.** SEC-07, SEC-10; SNFR-005; SFR-003, SFR-025.

Shall. Claims data, evidence, prompts, audit events, user acceptance testing records and exports shall be retained and destroyed according to an approved schedule, subject to an approved legal hold process.

Measurement method. Inspection of retention configuration against the approved schedule; destruction evidence sampling; legal-hold override testing.

Threshold or target. TBD — REQUIRES VALIDATION. No retention period exists for any data class. Periods must be set per class by the Compliance and Regulatory Owner with the Data Owner/Steward, informed by applicable United Arab Emirates health-sector obligations, before requirements-baseline approval.

Target approval state. TBD — requires validation.

Environment. All environments holding the affected data classes.

Evidence required. Approved retention and destruction schedule; legal-hold procedure; destruction evidence.

Owner role. Compliance and Regulatory Owner.

Failure or exception treatment. Absent an approved schedule, no data may be destroyed. Indefinite retention by default is itself a compliance exposure and is not a safe interim position.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**.

Open items. Depends on jurisdiction confirmation under DEP-02 and decision D-BRD-06, which remains Proposed. Retention of stamped result envelopes under SFR-003 is unresolved here.

SNFR-020 — Data quality

 **Derived requirement — no source NFR exists.** Data quality is a mandated quality attribute with no entry in the NFR catalogue, although dependency DEP-03 refers to approved quality thresholds. Recorded here so the gap stays visible; not an approved requirement until added to the NFR page under Change Control.

Source. None in the NFR catalogue. Derived from DEP-03, BR-06 and FR-13. **Related.** SFR-032, SFR-033; UC-01; UAT-09.

Shall. The system shall reject a candidate extract that fails approved data-quality thresholds, and shall report the failing measure rather than accepting degraded data.

Measurement method. Execution of the approved quality checks against a candidate extract; verification that a deliberately degraded extract is rejected.

Threshold or target. TBD — REQUIRES VALIDATION. Completeness, validity, uniqueness, timeliness and consistency thresholds do not exist. They must be set by the Data Owner/Steward before UAT entry. No threshold is invented here.

Target approval state. TBD — requires validation.

Environment. Test and pre-production onboarding.

Evidence required. Approved quality thresholds; check results per generation; rejection evidence for a degraded extract.

Owner role. Data Owner/Steward.

Failure or exception treatment. A failing extract is rejected and rolled back under SNFR-006. Accepting a known-degraded extract requires a recorded decision, not a silent override.

Implementation status. Catalogue: **Partly implemented** · RTM: **Designed / not implemented**. Schema, enumeration and window validation are implemented; quality thresholds beyond structural validity are not.

Open items. The catalogue gap itself; DEP-03 outstanding; field-level semantics unresolved under I-06.

9.8 Accessibility and usability

SNFR-015 — Accessibility

Source. NFR-15 (Must). **Related.** FR-07, FR-08, FR-09, FR-15 to FR-18; SFR-017 to SFR-025, SFR-038 to SFR-042; UC-02 to UC-09; UAT-10, UAT-11.

Shall. The pre-production user interfaces shall support keyboard-only operation, visible focus, meaningful labels, accessible status and error communication, sufficient colour contrast, and

the applicable WCAG 2.2 Level AA success criteria.

Measurement method. Automated accessibility scanning plus manual audit, including a keyboard-only walkthrough of every primary workflow and verification that status and error states are announced.

Threshold or target. No critical accessibility violation. All primary workflows completable using a keyboard alone. Applicable WCAG 2.2 Level AA criteria met. These are stated on the NFR page and are adopted unchanged; the definition of "critical" requires confirmation.

Target approval state. Standard named and adopted; **audit not executed.** No implemented interface exists to audit.

Environment. Pre-production once interfaces exist.

Evidence required. Automated scan results; manual audit report; keyboard walkthrough evidence per primary workflow; retained remediation record.

Owner role. Product Owner.

Failure or exception treatment. A critical violation blocks UAT entry. Accessibility of the grounding-blocked state under SFR-011 and the role-denied state under SFR-012 is explicitly in scope — those are the states most often omitted from an accessibility pass, and they are the states that carry the safety message.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**.

Open items. DEP-05 outstanding. The severity classification for accessibility findings needs agreement so that "no critical violation" is measurable.

SNFR-016 — Usability: viewport and operability

Source. NFR-16 (Should). **Related.** FR-07, FR-08, FR-09; SFR-017 to SFR-025; UAT-10, UAT-11.

Shall. Primary workflows shall remain readable and operable at the approved reference viewport and at the approved minimum supported viewport, without clipped controls, hidden required information or blocked actions.

Measurement method. Execution of executive, analyst and audit primary workflows at both viewports, checking for clipping, hidden required information and blocked actions.

Threshold or target. Reference viewport 1536 by 1024, as defined in the approved user-interface style specification. **Minimum supported viewport is TBD — requires validation;** the NFR page states that it requires approval and no value is invented here. Target: all three primary

workflows complete at both viewports with no blocked action and no loss of required information.

Target approval state. Reference viewport defined in design evidence; minimum supported viewport TBD — REQUIRES VALIDATION by the Product Owner before UAT entry.

Environment. Pre-production once interfaces exist.

Evidence required. Visual comparison at the approved reference size; workflow completion evidence at the approved minimum viewport; retained viewport policy.

Owner role. Product Owner.

Failure or exception treatment. A blocked action at a supported viewport is a defect, not a presentation preference.

Implementation status. Catalogue: **Designed / not implemented** · RTM: **Designed / not implemented**.

Open items. Mock-up values are illustrative under I-05 and must not be hardcoded. Whether the minimum viewport includes tablet or mobile form factors is undecided and materially affects design effort.

9.9 Unapproved target register

Consolidated view of every target in this section that is not approved. This register exists so that no reader has to infer approval state from prose, and so that the outstanding decisions can be worked as a list.

SNFR	Target	State	Owner role	Required by gate
SNFR-005	Audit search response and retention period	TBD — requires validation	Compliance and Regulatory Owner	Before UAT entry
SNFR-007	99.5% during agreed support hours; support window undefined	Proposed — not approved	Platform/Support Owner	Before requirements-baseline approval
SNFR-008	Verified in-region routing and	TBD — requires	Information Security and	Before UAT entry

	residency evidence	validation	Privacy Owner	
SNFR-009	p95 $\leq$ 15 s including tools; question set and concurrency undefined	Proposed — not approved	Platform/Support Owner	Before requirements-baseline approval
SNFR-010	p95 $\leq$ 2 s; covered route set undefined	Proposed — not approved	Platform/Support Owner	Before requirements-baseline approval
SNFR-011	50 named users / 10 concurrent; extract size and growth undefined	Proposed — not approved	Platform/Support Owner	Before requirements-baseline approval
SNFR-012	UAE business hours; severity, escalation and incident SLAs undefined	Proposed — not approved	Platform/Support Owner	Before requirements-baseline approval
SNFR-013	RPO 24 h / RTO 8 h; restore test frequency undefined	Proposed — not approved	Platform/Support Owner	Before requirements-baseline approval
SNFR-014	Retention periods per data class; legal-hold process	TBD — requires validation	Compliance and Regulatory Owner	Before requirements-baseline approval
SNFR-015	Severity classification defining "critical violation"	TBD — requires validation	Product Owner	Before UAT entry
SNFR-016	Minimum supported viewport	TBD — requires	Product Owner	Before UAT entry

		validation		
SNFR-018	Alert thresholds, routing and acknowledgement times	TBD — requires validation	Platform/Support Owner	Before UAT entry
SNFR-019	Formalised peer review, migration/rollback and approval procedure	TBD — requires validation	Engineering/Delivery Lead	Before requirements-baseline approval
SNFR-020	Data-quality thresholds per measure	TBD — requires validation	Data Owner/Steward	Before UAT entry

Summary. Of twenty non-functional system requirements, five are evidenced at the backend (SNFR-001, SNFR-002, SNFR-006, SNFR-017 and, in part, SNFR-019), seven carry a proposed but unapproved figure, and eleven carry at least one target that does not exist at all. No requirement in this section carries an approved service level. Assumption A-04 covers the proposed operational figures collectively and remains open.

9.10 Coverage statement

NFR-01 to NFR-16 are each represented: NFR-01 → SNFR-001; NFR-02 → SNFR-002; NFR-03 → SNFR-003; NFR-04 → SNFR-004; NFR-05 → SNFR-005; NFR-06 → SNFR-006; NFR-07 → SNFR-007; NFR-08 → SNFR-008; NFR-09 → SNFR-009; NFR-10 → SNFR-010; NFR-11 → SNFR-011; NFR-12 → SNFR-012; NFR-13 → SNFR-013; NFR-14 → SNFR-014; NFR-15 → SNFR-015; NFR-16 → SNFR-016. The NFR count remains 16 and no NFR wording has been altered.

Mandated quality attributes map as follows: grounding and correctness SNFR-001; deterministic behaviour SNFR-002; role-based access control SNFR-003; auditability SNFR-005; rollback and recoverability SNFR-006 and SNFR-013; portability SNFR-017; performance SNFR-009 and SNFR-010; availability SNFR-007; scalability SNFR-011; observability SNFR-018; supportability SNFR-012; maintainability SNFR-019; data quality SNFR-020; disaster recovery and continuity SNFR-013; accessibility SNFR-015 under NFR-15; usability SNFR-016 under NFR-16. Privacy (SNFR-004) and residency (SNFR-008) are additionally covered from NFR-04 and NFR-08.